ONE MONO CHANNEL — DISCRETE CLASS AB / +/−35 V

REV A: FIRST SCHEMATIC FOR ENGINEERING REVIEW. 8 ohm target; 4 ohm capability needs SOA/thermal qualification.

INPUT

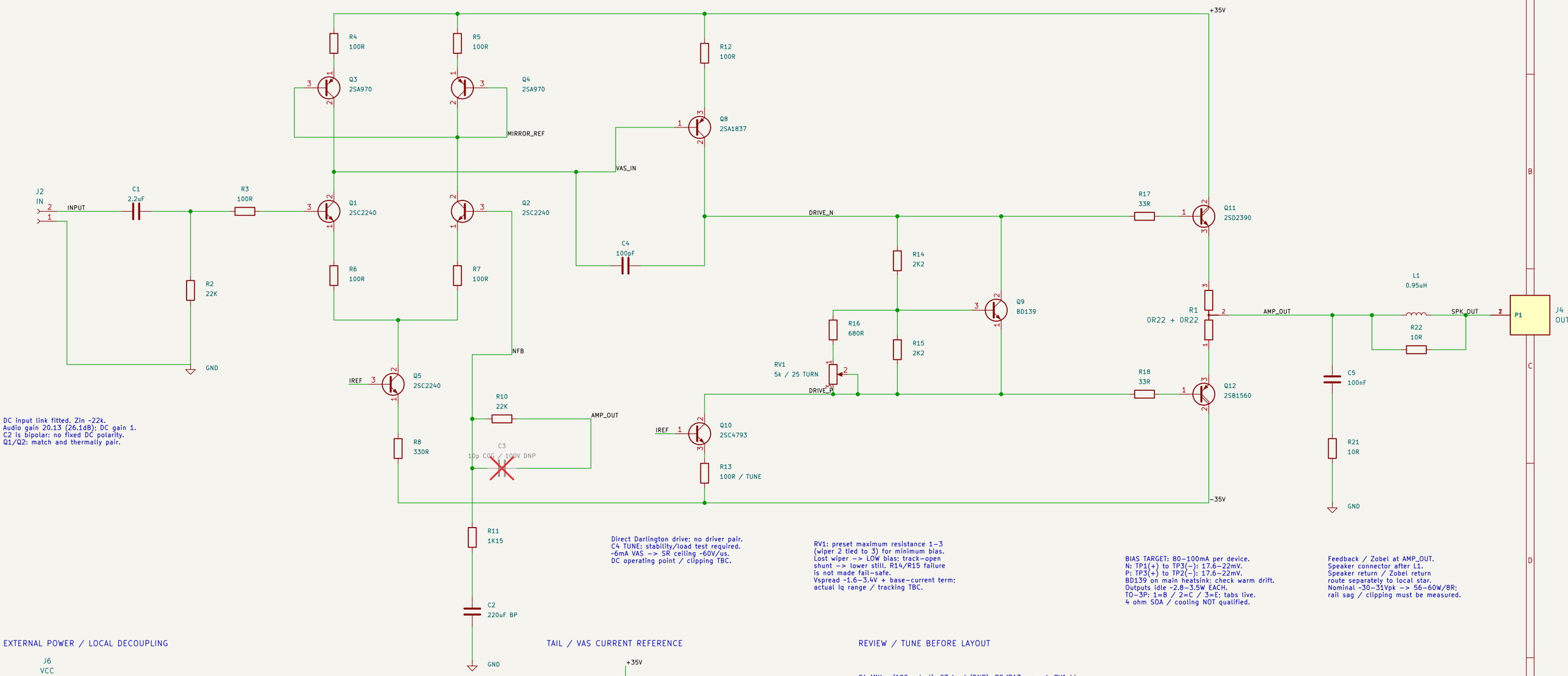
DIFFERENTIAL / MIRROR

VAS / CURRENT SINK

BIAS SPREADER

DARLINGTON OUTPUT

OUTPUT NETWORK



DC input link fitted. Zin ~22k.
Audio gain 20.13 (26.1dB); DC gain 1.
C2 is bipolar: no fixed DC polarity.
Q1/Q2: match and thermally pair.

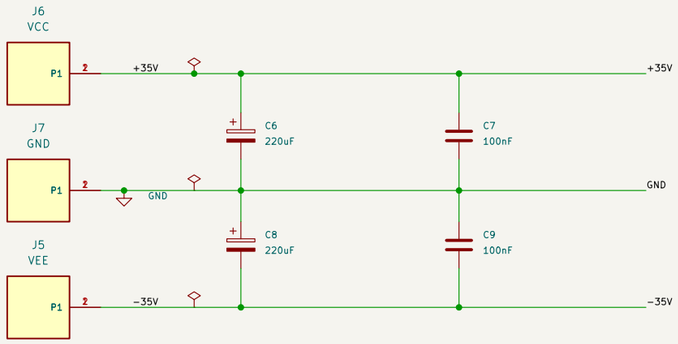
Direct Darlington drive; no driver pair.
C4 TUNE: stability/load test required.
~6mA VAS ~> SR ceiling ~60V/us.
DC operating point / clipping TBC.

RV1: preset maximum resistance 1–3
(wiper 2 tied to 3) for minimum bias.
Lost wiper ~> LOW bias; track-open
shunt ~> lower still. R14/R15 failure
is not made fail-safe.
Vspread ~1.6–3.4V + base-current term;
actual Iq range / tracking TBC.

BIAS TARGET: 80–100mA per device.
N: TP1(+) to TP3(−): 17.6–22mV.
P: TP3(+) to TP2(−): 17.6–22mV.
BD139 on main heatsink; check warm drift.
Outputs idle ~2.8–3.5W EACH.
TO-3P: 1=B / 2=C / 3=E; tabs live.
4 ohm SOA / cooling NOT qualified.

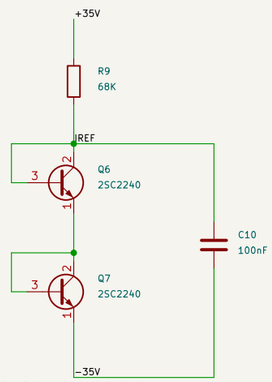
Feedback / Zobel at AMP_OUT.
Speaker connector after L1.
Speaker return / Zobel return
route separately to local star.
Nominal ~30–31Vpk ~> 56–60W/8R;
rail sag / clipping must be measured.

EXTERNAL POWER / LOCAL DECOUPLING



One electrical GND net. Local star/reference at J2 pin 2.
Input + feedback returns form quiet branches to that point.
Speaker, Zobel and decoupling returns use separate power paths.
Generic connector and passive footprints are TBC before layout.
J2/J3: select >=10A rated terminals; no series chosen yet.

TAIL / VAS CURRENT REFERENCE



IREF ~−33.8V. Tail ~1.8mA; VAS ~6mA.
R8/R13 TUNE: diode and VBE spread matter.
Q1/Q2 collectors ~+33V, emitters ~−0.7V.
Q5 ~60mW; front-end VCEO rating 120V.
Medium-power Q8/Q10 rated 230V.

REVIEW / TUNE BEFORE LAYOUT

C4 Miller (100p start), C3 lead (DNP), R8/R13 current, RV1 bias.
Check square-wave response with 8R / 4R / capacitive loads.
Check offset, startup, clipping recovery and heatsink bias tracking.
No DC protection, current limiting or thermal protection in Rev A.
At +/−35V a single pair is NOT signed off for sustained 4R use.
Worst resistive Class-B dissipation ~15.5W/device (8R),
~31W/device (4R), plus bias and reactive-load stress.
Small-signal leadforms and all exact passive/connector packages TBC.
BD139 symbol uses onsemi ECB pin map; verify any stocked ST part
independently (ST legacy internal schematic numbering conflicts).

Schematic only. Stability, bias and 4 ohm SOA require bench review.

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Title: DarlingtonAmp — Mono Class AB

Size: A3	Date: 2026-10-05	**Rev: A — REVIEW**
KiCad E.D.A. 10.0.5		Id: 1/1